Here each term is divided by 10. Hence, the next three terms are

$$a_4 = \frac{0.01}{10} = 0.001$$

$$a_5 = \frac{0.001}{10} = 0.0001$$

$$a_6 = \frac{0.0001}{10} = 0.00001$$

Example 2.20 Find the general term for the following sequences

(i) $3, 6, 9, \ldots$ (ii) $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \ldots$ (iii) $5, -25, 125, \ldots$

Solution (i) $3, 6, 9, \ldots$

Here the terms are multiples of 3. So the general term is

$a_n = 3n,$

(ii) $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \ldots$

$a_1 = \frac{1}{2};\ a_2 = \frac{2}{3};\ a_3 = \frac{3}{4}$

We see that the numerator of nth term is n, and the denominator is one more than the numerator. Hence, $a_n = \frac{n}{n+1}, n \in \mathbb{N}$

(iii) $5, -25, 125, \ldots$

The terms of the sequence have + and – sign alternatively and also they are in powers of 5.

So the general term $a_n = (-1)^{n+1}5^n, n \in \mathbb{N}$

Example 2.21 The general term of a sequence is defined as

$$a_n = \begin{cases} n(n+3); & n \in \mathbb{N} \text{ is odd} \\ n^2 + 1 \quad ; & n \in \mathbb{N} \text{ is even} \end{cases}$$

Find the eleventh and eighteenth terms.

Solution To find a_{11}, since 11 is odd, we put $n = 11$ in $a_n = n(n+3)$

Thus, the eleventh term $a_{11} = 11(11+3) = 154$.

To find a_{18}, since 18 is even, we put $n = 18$ in $a_n = n^2 + 1$

Thus, the eighteenth term $a_{18} = 18^2 + 1 = 325.$

Example 2.22 Find the first five terms of the following sequence.

$a_1 = 1,\ a_2 = 1, a_n = \frac{a_{n-1}}{a_{n-2} + 3};\ n \geq 3,\ n \in \mathbb{N}$

Solution The first two terms of this sequence are given by $a_1 = 1,\ a_2 = 1$. The third term a_3 depends on the first and second terms.

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